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Erkan Tairi

Education

2019–2024 **PhD in Computer Science**, *TU Wien*, Vienna, Austria
Supervisors: Matteo Maffei (TU Wien) and Daniel Slamanig (AIT Austrian Institute of Technology)
Thesis: Foundations of Adaptor Signatures for Distributed Ledger Protocols
`https://doi.org/10.34726/hss.2024.123264`

2016–2018 **MSc in Computer Science**, *Johannes Kepler University*, Linz, Austria

2011–2015 **BSc in Computer Science**, *University St. Paul the Apostle*, Ohrid, Macedonia

Professional Experience

Oct. 2025 - **Postdoctoral Researcher**, *UC Berkeley*, Berkeley, CA, USA
Host: Sanjam Garg

Sep. 2025 - **Research Scientist**, *Alloc Init*, Remote
Cryptographic protocol design and documentation

May 2024 - Sep. 2025 **Postdoctoral Researcher**, *ENS Paris*, Paris, France
Host: David Pointcheval

Feb. 2022 - Jan. 2024 **Research Engineer**, *MyPrivacy*, Vienna, Austria
Cryptographic protocol design and development

Internships and Visits

Feb. 2025 **Research Visit**, *King's College London*, London, England
Host: Martin Albrecht

Dec. 2022 **Research Visit**, *IMDEA Software Institute*, Madrid, Spain
Host: Dario Fiore

Nov. 2022 **Research Visit**, *ETH Zürich*, Zürich, Switzerland
Host: Dennis Hofheinz

Jun.-Sep. 2018 **Internship**, *AIT Austrian Institute of Technology*, Vienna, Austria
Supervisor: Daniel Slamanig

Oct. 2018 - Mar. 2019 **Internship**, *TU Wien*, Vienna, Austria
Supervisor: Matteo Maffei

Teaching and Supervision

2019–2023 **Teaching Assistant**, *TU Wien*, Vienna, Austria
Introduction to Cryptography, Privacy-Enhancing Cryptography, Cryptocurrencies, (graduate level)

Master Thesis Supervision
Aron Wussler (TU Wien) – Post-quantum Cryptography in OpenPGP
`https://doi.org/10.34726/hss.2023.106226`
Ngaffo Raissa Flora (University of Bamenda) – Simulation-extractable NIZK Proof Systems

Publications

Published Papers

Hardness of M-LWE with General Distributions and Applications to Leaky Variants. In PKC 2026.
Katharina Boudgoust, Corentin Jeudy, Erkan Tairi and Weiqiang Wen
`https://eprint.iacr.org/2025/1472`

On Verifiable Delay Functions from Time-lock Puzzles. In PKC 2026.
Hamza Abusalah, Karen Azari, Dario Fiore, Chethan Kamath and Erkan Tairi
`https://eprint.iacr.org/2025/1782`

Impossibility of VDFs in the ROM: The Complete Picture. In EUROCRYPT 2026.
Hamza Abusalah, Karen Azari, Chethan Kamath, Erkan Tairi and Maximilian von Consbruch
`https://eprint.iacr.org/2025/1773`

Lower Bounds for Lattice-based Compact Functional Encryption. In EUROCRYPT 2024.
Erkan Tairi and Akin Ünal
`https://doi.org/10.1007/978-3-031-58723-8_9`

(Inner-Product) Functional Encryption with Updatable Ciphertexts. In Journal of Cryptology.
Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and Erkan Tairi
`https://doi.org/10.1007/s00145-023-09486-y`

Optimizing 0-RTT Key Exchange with Full Forward Security. In ACM CCSW 2023.
Christian Göth, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks, Erkan Tairi and Alexander Zikulnig
`https://doi.org/10.1145/3605763.3625246`

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures. In ACM CCS 2023.
Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind
`https://doi.org/10.1145/3576915.3623149`

Foundations of Coin Mixing Services. In ACM CCS 2022.
Noemi Glaeser, Matteo Maffei, Giulio Malavolta, Pedro Moreno-Sanchez, Erkan Tairi and Sri AravindaKrishnan Thyagarajan
`https://doi.org/10.1145/3548606.3560637`

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs. In IEEE S&P 2021.
Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei
`https://doi.org/10.1109/sp40001.2021.00111`

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments. In FC 2021.
Erkan Tairi, Pedro Moreno-Sanchez and Matteo Maffei
`https://doi.org/10.1007/978-3-662-64331-0_7`

Updatable Signatures and Message Authentication Codes. In PKC 2021.
Valerio Cini, Sebastian Ramacher, Daniel Slamanig, Christoph Striecks and Erkan Tairi
`https://doi.org/10.1007/978-3-030-75245-3_25`

Preprints (Under Submission)

Registered Functional Encryption for Pseudorandom Functionalities from Lattices: Registered ABE for Unbounded Depth Circuits and Turing Machines, and More.
Tapas Pal, Robert Schädlich and Erkan Tairi
`https://eprint.iacr.org/2025/967`

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE.
Valerio Cini and Erkan Tairi
`https://eprint.iacr.org/2026/062`

Ciphertext-Updatable Attribute-based and Predicate Encryption from Lattices.
Robert Schädlich, Linda Scheu-Hachtel, Erkan Tairi and Yuejun Wang

LeOPaRd: Towards Practical Post-Quantum Oblivious PRFs via Interactive Lattice Problems.
Muhammed F. Esgin, Ron Steinfeld, Erkan Tairi and Jie Xu
https://eprint.iacr.org/2024/1615

Foundations of Verifiably Encrypted (Blind) Signatures.
Diego Castejon-Molina, Erkan Tairi, Dimitrios Vasilopoulos and Pedro Moreno-Sanchez

Adaptor Signatures Meet BLS: Efficient Blockchain Applications with Unique Adaptor Signatures.
Javier Gomez-Martinez, Erkan Tairi, Pedro Moreno-Sanchez and Clara Schneidewind

Lighthouse: Single-Server Secure Aggregation with O(1) Server-Committee Communication.
Sanjam Garg, Alireza Kavousi, Dimitris Kolonelos, Erkan Tairi, and Zhipeng Wang

Professional Activities

Program Committee

ACM CCS 2025-2026, ACM AsiaCCS 2026, FC 2025-2026, IACR Communications in Cryptology 2025-2026, CVC 2025

External Reviewer

ACISP 2024; ACM CCS 2021-2023; ACM AFT 2022; ACNS 2024-2025; APKC 2021-2022; Asiacrypt 2021, 2024-2025; CANS 2022; Crypto 2023-2025; Eurocrypt 2023, 2025-2026; FC 2021-2024; IEEE S&P 2024; IWSEC 2021-2023; PKC 2026; ProvSec 2020-2023

Administration and Organization

- Co-organizer of ViSP Cryptography Research Meetup
- Member of NDSS 2023 Student Support Committee

Awards and Grants

Erwin Schrödinger Fellowship (by Austrian Science Fund)
https://doi.org/10.55776/J4879

(Selected) Presentations and Invited Talks

(Fine-Grained) Unbounded Inner-Product Functional Encryption from LWE
Crypto Day at Télécom Paris. Jun. 2025
https://shorturl.at/DrJEr
Guest talk at King's College London Cybersecurity Group. Feb. 2025

LedgerLocks: A Security Framework for Blockchain Protocols Based on Adaptor Signatures
IOG Seminar. Nov. 2023

(Inner-Product) Functional Encryption with Updatable Ciphertexts
PICOCRYPT Seminar at IMDEA Software Institute. Dec. 2022
https://shorturl.at/5RP3o

A2L: Anonymous Atomic Locks for Scalability in Payment Channel Hubs
IEEE Symposium on Security and Privacy. May 2021
https://www.youtube.com/watch?v=RPj8efQteyg

Post-Quantum Adaptor Signature for Privacy-Preserving Off-Chain Payments
Financial Cryptography and Data Security 2021. Mar. 2021
https://www.youtube.com/watch?v=BqMDnWFoGL0
Decrypto Seminar. Dec. 2020

Updatable Signatures and Message Authentication Codes
Conference PKC 2021. May 2021

https://www.youtube.com/watch?v=RqAlXixDZ1M
Young Researcher Crypto Seminar. May 2021

Computer Skills

Programming	C, C++, Rust *(advanced)*, C#, Python, Shell *(average)*, Java, Go *(basics)*
Web	HTML, CSS, JavaScript (React, Node.js), ASP.NET Web *(advanced)*
Software	Visual Studio, SageMath, Magma, Git, LaTeX
Other	Microsoft Azure, Amazon AWS (cloud computing)

Languages

Macedonian	Native	
English	Fluent	*professional working proficiency*
Turkish	Fluent	*professional working proficiency*
French	Intermediate	*elementary proficiency*
German	Intermediate	*elementary proficiency*
Albanian	Intermediate	*elementary proficiency*

Referees

Sanjam Garg	*sanjamg@berkeley.edu*
David Pointcheval	*david.pointcheval@ens.fr*
Daniel Slamanig	*daniel.slamanig@unibw.de*
Matteo Maffei	*matteo.maffei@tuwien.ac.at*